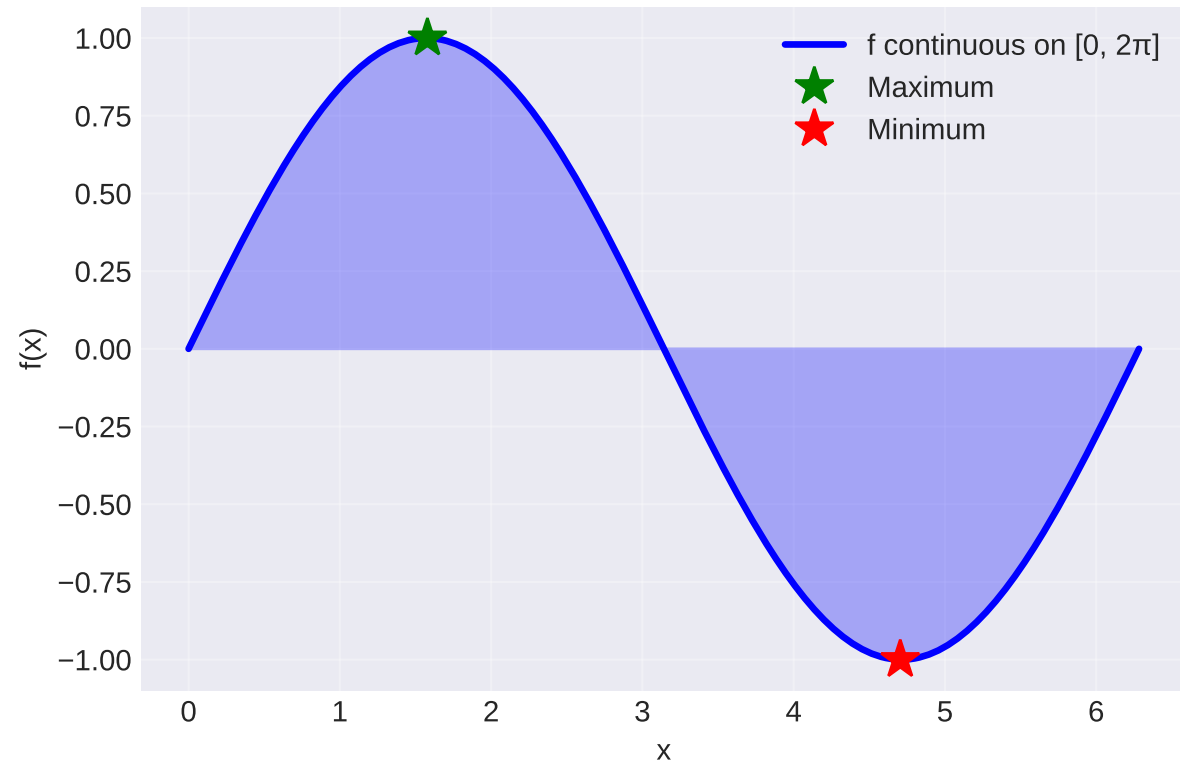


Continuous Functions on Compact Sets

Continuous on Compact Set
(attains max and min)



Key Theorem: Continuous Image of Compact Set

If K is compact and $f: K \rightarrow \mathbb{R}$ is continuous, then $f(K)$ is compact.

Consequences:

- f is bounded on K
- f attains its maximum and minimum
- f is uniformly continuous on K

Example:

$f(x) = \sin(x)$ on $[0, 2\pi]$:

- Bounded: $f(x) \in [-1, 1]$
- Max: 1 (at $x = \pi/2$)
- Min: -1 (at $x = 3\pi/2$)
- Uniformly continuous